Neural Networks - Answer Context & Reference

Topic: Neural Networks

About This Reference

This document contains the complete answer context for all 20 Neural Networks practice questions. Each entry includes the correct answer and a detailed explanation grounded in core Deep Learning theory. Use this as a study reference after attempting the question paper independently.

Q1. (Multiple Choice) [Easy] - What is a Neural Network inspired by?

Answer: The human brain and biological neurons

Artificial Neural Networks (ANN) are inspired by how biological neurons in the human brain communicate. Just as neurons pass signals through synapses, artificial neurons pass weighted signals through layers.

Q2. (Multiple Choice) [Easy] - Which of the following is NOT a layer type in a basic Neural Network?

Answer: Processing Layer

A basic Neural Network has three layer types: Input Layer (receives features), Hidden Layer(s) (processes and transforms data), and Output Layer (produces final prediction). There is no 'Processing Layer' - that is not standard terminology.

Q3. (Multiple Choice) [Easy] - What is the role of an Activation Function in a Neural Network?

Answer: To introduce non-linearity so the network can learn complex patterns

Without activation functions, a neural network would just be a linear equation no matter how many layers. Activation functions like ReLU, Sigmoid, and Tanh add non-linearity, allowing the network to learn complex, non-linear patterns.

Q4. (Multiple Choice) [Medium] - What is the ReLU activation function formula?

Answer: $f(x) = \max(0, x)$

ReLU (Rectified Linear Unit) = $\max(0, x)$. It outputs x if $x > 0$, and 0 if $x <= 0$. ReLU is computationally efficient and avoids the vanishing gradient problem.

Q5. (Multiple Choice) [Medium] - What is Backpropagation in Neural Networks?

Answer: Algorithm to update weights by propagating error backwards from output to input

Backpropagation calculates the gradient of the loss function with respect to each weight using the chain rule. These gradients are used by Gradient Descent to update weights - this is how the neural network learns.

Q6. (Multiple Choice) [Medium] - What is a Convolutional Neural Network (CNN) primarily used for?

Answer: Image and spatial data processing

CNNs use convolutional layers to automatically detect features in images - edges, shapes, textures. They are the standard choice for medical image analysis, X-ray classification, and tumour detection.

Q7. (Multiple Choice) [Medium] - What problem does the Dropout regularization technique solve?

Answer: Overfitting

Dropout randomly deactivates a fraction of neurons during each training iteration, preventing the network from over-relying on specific neurons and forcing it to learn more robust features.

Q8. (Multiple Choice) [Hard] - What is the Vanishing Gradient problem?

Answer: When gradients become extremely small during backpropagation, making early layers learn very slowly

In deep networks with sigmoid/tanh activations, gradients shrink during backpropagation and become tiny near input layers. ReLU and batch normalization help solve this.

Q9. (Multiple Choice) [Hard] - In CNN architecture, what does a Pooling Layer do?

Answer: Reduces the spatial dimensions of the feature map while retaining important features

Pooling (Max or Average) reduces width and height of feature maps, lowering computational cost and controlling overfitting. Max Pooling takes the maximum value in each window, preserving prominent features.

Q10. (Multiple Choice) [Hard] - What is the difference between a shallow neural network and a deep neural network?

Answer: Shallow has 1 hidden layer; Deep has 2 or more hidden layers

Deep Learning refers to neural networks with multiple hidden layers. More layers allow learning increasingly abstract representations - from edges to shapes to objects in image recognition tasks.

Q11. (True / False) [Easy] - A Neural Network with zero hidden layers is equivalent to Logistic Regression.

Answer: True

A neural network with no hidden layers and a sigmoid output is mathematically equivalent to Logistic Regression. Hidden layers are what give neural networks their extra power.

Q12. (True / False) [Medium] - Increasing the number of layers always improves a Neural Network's performance.

Answer: False

More layers can lead to overfitting, vanishing gradients, and slower training without better accuracy. The right depth depends on the problem and data size.

Q13. (True / False) [Medium] - CNNs are the standard architecture for medical image classification tasks.

Answer: True

CNNs automatically learn spatial features through convolution. They are used in chest X-ray classification, tumour detection, retinal disease diagnosis, and skin lesion classification.

Q14. (True / False) [Hard] - Batch Normalization speeds up training by normalizing inputs to each layer.

Answer: True

Batch Normalization normalizes layer inputs to mean=0 and variance=1 within each mini-batch, reducing internal covariate shift and allowing higher learning rates.

Q15. (Short Answer) [Easy] - Explain how a Neural Network learns with a simple analogy.

Answer: A Neural Network learns like a student studying for an exam. Step 1 - Forward Pass: student gives an answer (prediction). Step 2 - Calculate Error: teacher marks it right or wrong (loss function). Step 3 - Backpropagation: student understands which concepts were wrong (gradient flows back). Step 4 - Weight Update: student studies harder on weak areas (Gradient Descent updates weights). Step 5 - Repeat for many examples (epochs) until performance improves.

This forward-backward learning cycle is called an epoch. Multiple epochs on training data gradually reduce the error.

Q16. (Short Answer) [Medium] - What are the main components of a CNN architecture? Explain each briefly.

Answer: 1) Input Layer - receives raw image as pixel matrix. 2) Convolutional Layer - applies filters to detect features (edges, textures). 3) Activation (ReLU) - adds non-linearity. 4) Pooling Layer - reduces spatial dimensions. 5) Flatten - converts 2D maps to 1D vector. 6) Fully Connected Layer - combines all features. 7) Output Layer - final classification (Softmax or Sigmoid).

Understanding CNN architecture is essential for medical imaging tasks like X-ray diagnosis and tumour detection.

Q17. (Short Answer) [Hard] - Explain Transfer Learning and why it is especially useful in biomedical AI.

Answer: Transfer Learning uses a pre-trained network (trained on millions of images) and fine-tunes it on a smaller domain-specific dataset. Useful in biomedical AI because: 1) Medical labelled data is scarce and expensive to annotate. 2) Pre-trained features (edges, textures) transfer well to medical images. 3) Fine-tuning is much faster than training from scratch. Example: ResNet or VGG16 trained on ImageNet fine-tuned for chest X-ray classification.

Transfer learning is why modern medical AI systems can achieve high accuracy with relatively small datasets.

Q18. (Multiple Choice) [Easy] - What activation function is used in the output layer for binary classification?

Answer: Sigmoid

Sigmoid outputs a value between 0 and 1, interpreted as a probability for binary classification. For multi-class classification, Softmax is used instead.

Q19. (Multiple Choice) [Medium] - What is an Epoch in Neural Network training?

Answer: One complete pass through the entire training dataset

An epoch is one complete pass through all training samples. Training typically requires many epochs (50-500+) until the model converges. Too many epochs leads to overfitting.

Q20. (Short Answer) [Hard] - How does a CNN detect diseases in medical images? Explain with reference to chest X-ray analysis.

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Answer: 1) Input - Chest X-ray image as pixel matrix (e.g. 224x224). 2) First Conv Layers - detect low-level features: edges, bone outlines. 3) Middle Layers - detect medium-level features: rib shapes, lung boundaries. 4) Deep Layers - detect high-level features: consolidation (pneumonia), nodules (tumour), effusion signs. 5) Fully Connected - combines all features. 6) Output - probability scores per class (e.g. Normal=0.12, Pneumonia=0.78). 7) Prediction - highest probability class selected.

Each CNN layer learns increasingly complex features automatically - no manual feature engineering needed.